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Exploring interface dynamics in plastic bonded-explosives: synthesis, characterization, and neutron reflectometry

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ABSTRACT

Understanding the interfacial properties of high energy explosives blended with polymer binders is important for optimizing material performance and predicting aging. In the case of PBX 9502, direct measurements of the TATB – Kel-F 800 interface have been limited by the lack of high-fidelity model systems. This paper reports the fabrication of a new thin film, model TATB sample to enable neutron reflectometry measurements of this interface. The fabricated TATB films were characterized using X-ray diffraction, scanning electron microscopy, and atomic force microscopy. The film deposits in the same polymorph as the TATB crystallites in PBX 9502, but in the form of highly textured nanopillars that are ~200 nm wide. The buried interface of the TATB film (i.e., bottoms of the pillars) has an rms roughness of ~3 nm and is the (001) crystal face. The TATB films retain their features when deposited on Kel-F 800 thin films, enabling the formation of layered PBX 9502 model samples. Neutron reflectometry measurements showed increasing roughness of the TATB – Kel-F 800 interface with thermal aging, consistent with void formation at this interface.

KEYWORDS

Neutron reflectometry; PBX 9502; TATB; thermal aging; thin film

Introduction

PBX 9502 is a granular composite of 95 wt% 2,4,6-triamino-1,3,5-trinitrobenzene (TATB) crystals within a Kel-F 800 polymer binder. PBX 9502 is formulated by adding a solution of Kel-F to a slurry of TATB particles, then precipitating the Kel-F onto the surface of the TATB particles by solvent

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crashing (Hallam 1976). Agglomerations of coated TATB particles, called prills, are dried and then pressed under heat and pressure ($\sim 100^\circ\text{C}$ and ~ 0.1 GPa) to form a charge (Olinger 2005; Schwarz et al. 2013; Willey et al. 2009). PBX 9502 has a complex microstructure that significantly changes throughout its manufacturing process. The focus of this paper is to develop new model systems to study the evolution of the TATB – polymer binder interface. We first describe the physical properties of Kel-F and TATB followed by the properties of PBX 9502 explosive charges and the TATB – Kel-F interface.

Kel-F 800 is a fluoropolymer made as a statistical copolymer of the monomers chlorotrifluoroethylene (CTFE) and vinylidene fluoride (VF_2), with ~ 0.25 molar fraction VF_2 (Cady and Caley 1977; Willey, DePiero, and Hoffman 2009). It has a low degree of crystallinity ($< \sim 15$ wt%) which varies with thermal history (Cady and Caley 1977; Hoffman, Matthews, and Cesar 1989).

TATB is an insensitive high explosive (HE) composed of a benzene ring with alternating amino and nitro groups. TATB forms several different crystalline lattices (Kolb and Rizzo 1979). However, only one crystal arrangement is relevant to PBX 9502: a triclinic $P1$ lattice. This unit cell contains two TATB molecules and has lattice parameters $a = 9.010 \text{ \AA}$, $b = 9.028 \text{ \AA}$, $c = 6.812 \text{ \AA}$, $\alpha = 108.59^\circ$, $\beta = 91.82^\circ$, and $\gamma = 119.97^\circ$ (Cady and Larson 1965). In this crystal structure, the benzene rings lie flat to form single-molecule thick, planar, graphite-like sheets that stack on top of one another in the $[001]$ direction. Within the sheets, there is strong hydrogen bonding between the molecules (Cady and Larson 1965), but between the sheets there are only weaker $\pi - \pi$ stacking and Van der Waals interactions (Zhang et al. 2023). TATB crystals for PBX 9502 are synthesized using dry-amination.¹ This process results in a dispersion of small single crystals or crystallites. The crystallites are high aspect ratio and plate-shaped (i.e., platelets), where the face of the plate is the (001) plane (Cady 1992; Luscher et al. 2017; Nandi et al. 2007; Yeager et al. 2016). The degree to which the crystallites are plate-shaped can be significantly enhanced or mitigated depending on the synthesis route. Each crystallite has a significant volume fraction of large voids, with SEM and computed tomography studies describing their appearance as “Swiss cheese” or “worm-holes” (Cady 1992; Patterson et al. 2022; Skidmore, Phillips, and Crane 1997). Ignoring these void contributions, the density of solid material is typically $\sim 99\%$ of the theoretical maximum density (TMD) of a TATB single crystal for the $P1$ lattice ($\text{TMD}_{\text{TATB}} = 1.937 \text{ g cm}^{-3}$) (Hoffman and Aaron 2010). These crystallites will aggregate and nucleate on one another to form larger particles of TATB. The aggregate particles range in size from around one to several hundred microns (Luscher et al. 2017; Nandi et al. 2007; Rizzo, Humphrey, and Kolb 1981). The crystals making up each particle tend to organize in layered sheets, described as a “stack of cards” or “filo dough” morphology (Luscher et al. 2017; Schwarz et al. 2013). The agglomerates can be dried into HE powder or formed into prills as described above.

HE samples can be pressed from TATB powder to make a neat-TATB charge or from Kel-F coated prills to make a PBX 9502 charge. The Kel-F binder likely has minimal influence on the deformation of TATB crystallites and particles while pressing because it has a small volume fraction and processing occurs above Kel-F's melting temperature. This means that the final pressed microstructure of TATB is largely the same in both neat-TATB and PBX 9502 charges. Before pressing, the tamped powder contains a random distribution of TATB platelet directions (Yeager et al. 2016). During pressing, these TATB platelets and particles deform primarily by shearing and bending, greatly reducing the void volume (Patterson et al. 2022; Schwarz et al. 2013; Yeager et al. 2016). The resultant charge is highly textured, where the TATB crystallites have re-oriented such that their (001) faces are perpendicular to the direction of maximum compressive strain. The high degree of alignment between the TATB crystallites primarily results from the plate-like aspect ratio and the weak inter-sheet Van der Waals interactions that cause TATB to have only one cleavage plane, the (001) face (Luscher et al. 2017; Schwarz et al. 2013; Yeager et al. 2016). The final pressed charges have elastically stressed layers of TATB platelets forming a "bent stack of cards" morphology (Luscher et al. 2017; Schwarz et al. 2013; Skidmore, Butler, and Sandoval 2003; Yeager et al. 2016). Typical densities for pressed neat-TATB charges are approximately $1.80\text{--}1.90\text{ g cm}^{-3}$, or $\sim 93\text{--}98\%$ of TMD for TATB (Daily et al. 2013; Gibbs, Popolato, and Baytos 1980; Kolb and Rizzo 1979; Luscher et al. 2017). Typical densities for isostatically pressed PBX 9502 charges are approximately $1.85\text{ to }1.91\text{ g cm}^{-3}$ or $95\text{--}98.5\%$ of TMD ($\text{TMD}_{\text{PBX 9502}} = 1.94\text{ g cm}^{-3}$) (Olinger 2005; Yeager et al. 2020).

The Kel-F binder significantly influences the void microstructure in TATB-based PBX charges. Void growth is undesirable primarily because shaped charge performance is dependent on final density and dimensional stability, and changes in void volume can affect detonation characteristics (Patterson et al. 2022). Measurements show PBX 9502 having significantly less void volume than a neat-TATB charge (Willey et al. 2006). This is because Kel-F under high heat and pressure can flow into unfilled areas during the pressing process. The voids that are left in PBX 9502 are a mixture of isolated voids and interconnected cracks (Mang and Hjelm 2021; Yeager et al. 2022). Similar to the TATB sheets, these voids have a preferred orientation. The larger dimensions of the voids, i.e., the length of the cracks, lie in the plane perpendicular to the direction of maximum compressive strain (same as the TATB sheet orientation) (Mang and Hjelm 2021). In the pressed cyclotetramethylene-tetranitramine (HMX)-based PBX 9501, voids are primarily located at the HE interfaces and there is almost no void space within the bulk of the polymer binder (Mang and Hjelm 2011). This appears to be the case for PBX 9502 as well (Patterson et al. 2022).

The Kel-F binder also mitigates some thermomechanical issues, the most important of which is ratchet growth caused by the extreme thermal expansion anisotropy of TATB crystals (Patterson et al. 2022; Thompson et al. 2015; Thompson, Woznick, and DeLuca 2019). During thermally driven ratchet growth, voids that are present grow larger, longer, and more interconnected, causing the charge to expand and decrease in density (Yeager et al. 2022). Void formation is posited to primarily propagate along the HE-binder interfaces in PBXs (Rae et al. 2002; Yeager et al. 2010). Consequently, binder-related properties (i.e., adhesion, wetting, and glass transition temperature T_g) significantly affect ratchet growth and fracturing in TATB-based PBXs (He et al. 2015; Maiti et al. 2008; Rizzo, Humphrey, and Kolb 1981; Willey, DePiero, and Hoffman 2009). This has motivated work to understand and quantify the interfacial properties for TATB-based PBXs.

Previous work most often quantifies the HE – binder interface behavior using bulk properties. For example, interface optimization has improved stiffness, yield strength, and creep resistance for TATB-based and model PBX materials (Herman, Liu, et al. 2023; Lin et al. 2015; Zheng et al. 2024; Zhou et al. 2024). Other work has characterized the TATB – binder interface behavior using changes in microstructure (Willey et al. 2009). Some efforts have also aimed to directly measure and model interfacial properties for both TATB-based PBXs (Gee et al. 2007; Rivera and Matuszak 1983; Williamson et al. 2014; Williamson et al. 2017; Yeager et al. 2010) and for PBXs in general (Herman, Bull, et al. 2023; Hoss et al. 2016; Kodali et al. 2023; Taylor et al. 2014; Williamson, Hamilton, and Jardine 2017; Yeager et al. 2012, 2013, 2018). However, direct measurements of the TATB-binder interface are challenging because most interfacial science measurements require large, smooth, flat surfaces, but appropriate TATB model samples are not available. TATB single crystals are far too small, while agglomerate particles have irregular geometries and void structures. Pressed TATB charges are also not sufficiently smooth, and TATB is too soft to polish – although a newly developed technique could change this for mm-scale areas (Yeager et al. 2022). As a result, studies of the TATB interface are lacking. HE thin films have previously been developed, using dip coating (Yeager et al. 2013, 2018) and vapor deposition (Forrest et al. 2021; Hoss et al. 2016; Knepper et al. 2011; Knepper et al. 2012; Olles et al. 2019; Tappan, Wixom, and Knepper 2014), that conform to the above requirements and still serve as mimetic systems. In particular, vapor deposition with appropriate choice of substrate has recently been developed as a means to produce reasonably smooth TATB films over macroscopic areas (Williamson et al. 2017). Building on this development, this work describes the fabrication and characterization of vapor deposited, thin film TATB model samples in order to develop an improved method to evaluate the TATB – binder interface. Smooth and large surface area thin films were produced and characterized. Methods for deposition of Kel-F layers

are also described to produce layered composites for neutron reflectivity measurements. In these measurements, an increase in roughness at the TATB – Kel-F 800 interface was detected as a function of modest thermal cycling.

Materials and Methods

Thin films of TATB (1–30 μm thick) were fabricated using physical vapor deposition in a custom high-vacuum chamber. TATB powder was loaded into an effusion cell deposition source, and the chamber was evacuated to a base pressure of $\sim 10^{-6}$ Torr. The effusion cell was heated to a maximum temperature of 240°C, at which point the TATB sublimed over a period of several hours. Substrates were positioned in a rotating sample holder above the effusion cell and in contact with a water-cooled copper block. A shadow mask was used to define the deposition geometry. TATB film thickness was measured using a Bruker Dektak XT stylus profilometer.

The X-ray diffraction (XRD) sample was made by vapor depositing a $\sim 24\ \mu\text{m}$ TATB thin film onto a 25 mm diameter round, polished silicon (Si) substrate. XRD scans were done using a Kratos Analytical Diffractometer (θ - 2θ scans, Cu K α X-ray source, scan range of $2\theta = 5^\circ$ – 50° , increments of 0.04° , 1 second integration time).

The scanning electron microscope (SEM) sample was made by depositing a $\sim 2\ \mu\text{m}$ TATB thin film onto a 25 mm diameter round, polished BK7 glass substrate. It was cross sectioned by scoring the substrate with a diamond scribe outside of the film deposition area and then fracturing. Before SEM measurements, the sample was coated in $\sim 2\ \text{nm}$ of iridium to mitigate surface charge build up. SEM images were taken using a Zeiss GeminiSEM 300 with 1.10 kV electrons and a 20 μm aperture width.

Atomic Force Microscopy (AFM) measurements focused on the smooth, buried interface of vapor deposited TATB films because the exposed TATB interface is much rougher (described later). To make an AFM sample, we adapted liftoff protocols using a sacrificial polymer film to access and expose the buried TATB interface (see SI [Section 3](#)) (Linder et al. 2005). 10 wt% polyacrylic acid in water (PAA, Polysciences Inc, $M_w \approx 50,000$) spin coated at 1000 rpm to a thickness of $\sim 0.7\ \mu\text{m}$ was used as the sacrificial layer (see SI [Section 2](#)). A $0.9\ \mu\text{m}$ TATB thin film was vapor deposited on top of the PAA film. The edges of the substrate were scored and the film coated substrate was placed in water to dissolve the PAA and free the TATB thin film. After liftoff, the TATB film was epoxied to a new substrate to support it during AFM measurements. AFM scans used a Bruker MMAFM-2 controlled by the NanoScope V531r1 software package. Measurements used Ted Pella TAP300Al-G-50 tips (force constant $\sim 40\ \text{N m}^{-1}$, tip radius $\sim 10\ \text{nm}$, resonant

frequency ~ 300 kHz) in tapping mode, 512×512 pixel scans, and line measurement speeds of 0.25–0.5 hz.

Three types of samples were made for neutron reflectometry: Kel-F calibration, TATB calibration, and model PBX 9502. A Kel-F calibration sample was made by spin coating Kel-F 800 (3 M) on a polished Si wafer in an ethyl acetate vapor saturated environment. The Kel-F thin films had a free surface rms roughness of ~ 0.9 nm by AFM and ~ 0.5 nm by neutron reflectometry (see SI section 6.1). A TATB calibration sample was made by vapor depositing a ~ 30 μm TATB thin film directly onto a polished 25 mm diameter, 5 mm thick Si substrate. Model PBX 9502 samples were made in two steps. First, 3 wt% Kel-F 800 dissolved in ethyl acetate (Sigma Aldrich, HPLC grade, 99.8%) was spin coated at 1000 rpm for 5 minutes in a vapor saturated environment onto 25 mm diameter, 5 mm thick, polished Si substrates (see SI section 1). Then, a ~ 5 μm TATB thin film was vapor deposited onto the Kel-F surface. After being measured in their initial, as-fabricated states, the model PBX 9502 samples were subjected to thermal cycling using standard convection ovens (Quincy Labs, Inc. 20GC convection oven). The ovens had approximate ramp rates of 6°C min^{-1} for heating and 5°C min^{-1} for cooling.

Neutron Reflectometry (NR, see SI section 5) measurements were done on the Asterix time-of-flight (TOF) reflectometer (FP-11, ER-2) at the Manuel Lujan Jr. Neutron Scattering Center, Los Alamos National Laboratory. The beamline is a horizontal reflectometer with cold neutron wavelengths between approximately $4 < \lambda < 13$ Å. For the NR measurements on model PBX 9502 samples, detector images were collected at angles $0.5^\circ \leq \theta \leq 2.25^\circ$ covering a momentum transfer q range of $0.008 \leq q \leq 0.123$ Å $^{-1}$. Due to limited beamtime, the collected q range was truncated to $0.008 \leq q \leq 0.06$ Å $^{-1}$ for some measurements. Reflectivity $R(q)$, the ratio of reflected beam intensity to incident beam intensity, for each measurement was collected at 3 discrete angles (2 for truncated measurements). Exposure times ranged from 20 minutes at low angles to 4 hours at high angles. NR data was fitted using the software package MIRROR developed by W.A. Hamilton and J.B. Hayter. The MIRROR neutron reflectivity fitting routine is based on an iterative, dynamical method called “downhill simplex” (Nelder and Mead 1965). Further discussion of the MIRROR mathematical methods are featured in Mulder and Kuhl (2010). Simple layer models were used to fit the data, while the silicon substrate was fixed with a Scattering Length Density (SLD) of $\beta_{\text{Si}} = 2.08 \times 10^{-6}$ Å $^{-2}$. A separate oxide layer on silicon did not improve the quality of the fit and so was not included in our layer models. The Kel-F calibration film fits an SLD with an error range of 4.00 to 4.14×10^{-6} Å $^{-2}$ (see SI Section 6.1). The TATB calibration film had an SLD $\beta_{\text{TATB}} = 4.67 \pm 0.13 \times 10^{-6}$ Å $^{-2}$ calculated using the critical edge.

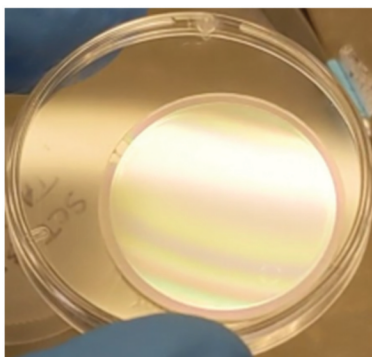


Figure 1. TATB film on a quartz substrate in a plastic petri dish. Prior to TATB vapor deposition, the quartz was spin coated with PAA so that the TATB was deposited onto a polymer film instead of directly onto the quartz substrate. The interference pattern is seen under the emission of a fluorescent tube light.

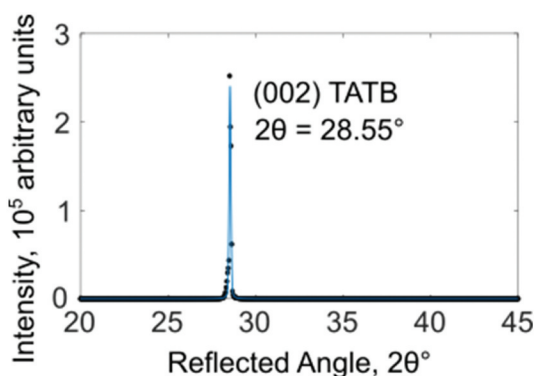


Figure 2. XRD of a vapor deposited thin TATB film on a Si substrate using Cu K α X-rays. There is a single TATB peak at $2\theta = 28.55^\circ$, which corresponds to a d-spacing of 3.12 Å and matches the (002) reflection.

Results

Characterization of Vapor Deposited TATB Films

Figure 1 shows a vapor deposited TATB thin film on a quartz substrate. The TATB is deposited on top of a PAA thin film. The wide, uniform interference fringes in Figure 1 show that the TATB film is flat and of mostly uniform thickness. The thin ring of non-uniform thickness at the outer edge of the sample is due to the PAA edge bead. From inspection under a microscope at 10X, 20X, and 40X, the as-deposited TATB films were highly continuous and largely defect-free with only a few point defects per film.

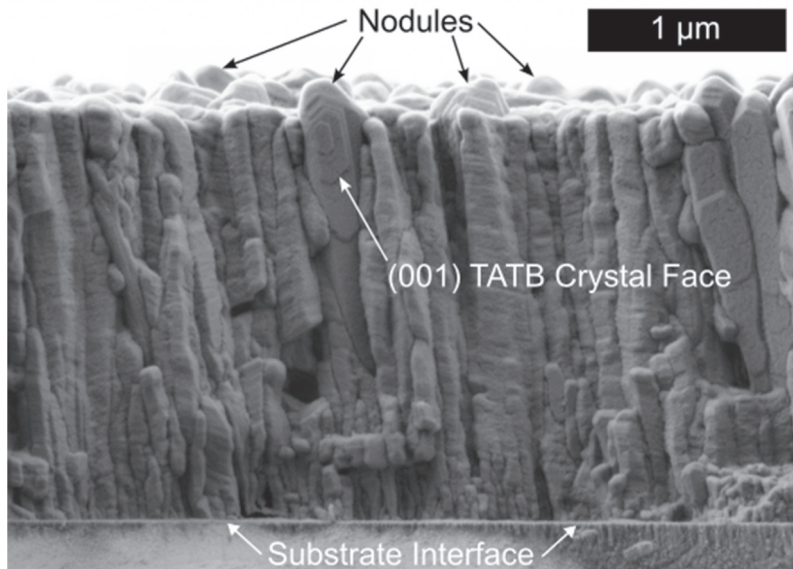
Figure 2 shows the XRD from a sample of TATB deposited onto an Si substrate. The strong TATB reflection indicates that the film has high crystallinity. The small FWHM of the peak (FWHM = 0.12°) demonstrates

that the TATB crystals have minimal lattice variation and that the crystalline domains are relatively large. Assuming a shape factor of 1, the in-plane Scherrer domain size is $\tau = 75$ nm (Smilgies 2009). The single TATB peak at $2\theta = 28.55^\circ$ corresponds to a d -spacing of 3.12 Å and matches the (002) reflection in the XRD pattern of typical triclinic crystalline TATB particles (Luscher et al. 2017; Schwarz et al. 2013; Yang et al. 2006). For XRD on a thin film, higher intensities for diffraction reflections indicate that the indexed crystallographic direction is more aligned perpendicular to the substrate. The fact that there is only one TATB diffraction peak signifies that almost all the crystal grains within the film are aligned in the same direction. The grains should all have their (001) planes (i.e., the graphite-like sheets of TATB molecules) laying parallel to the substrate. If there were other contributing reflections, peaks would especially be expected around $2\theta = 22^\circ$ and $2\theta = 43^\circ$ (Schwarz et al. 2013). Note that there is no information about whether the TATB grains have their a - or b -axes aligned.

Figure 3 shows SEM images of a ~ 2 μm TATB film vapor deposited directly onto an BK7 glass substrate. At the bottom of the cross-section image (Figure 3a) is the buried TATB interface, i.e., the TATB – substrate interface. This interface appears sharp and smooth. The bulk of the TATB film cross section is dominated by columnar structures. The columns show a high degree of vertical alignment, are 100–400 nm wide, and extend all or most of the way through the height of the film. The horizontal striations in most of the columns suggest that the plate-like (001) planes are laying parallel to the substrate in agreement with the XRD results. The faceted nature of all the columnar structures is indicative of a highly crystalline film. At the free surface the tops of these columns appear slightly rounded or domed when viewed in cross-section (Figure 3a) and form a tile-pattern extending across the area of the film when viewed top-down (Figure 3b).

The free surface of the TATB film has many oblong nodules that are 50–100 nm in height and are ~ 100 nm in diameter. The nodules can be seen in-profile in Figure 3(a), or top-down in Figure 3(b). The growth of nodule-like structures is a common problem for (comparatively) thick films grown using physical vapor deposition methods (Panjan et al. 2020). These nodules are distributed evenly and randomly across the free surface at a surface density of approximately 3 nodules/μm². At least some of the nodule defects are caused by the growth of TATB columns with the (001) crystal plane oriented perpendicular to the substrate instead of parallel. This is shown in Figure 3(a), where columns ending in nodules show the plate-like faces of the (001) plane facing out of the image. The nodules greatly increase the roughness of the free surface making it unusable for NR. Additional SEM images are provided in SI Section 4.

(a)



(b)

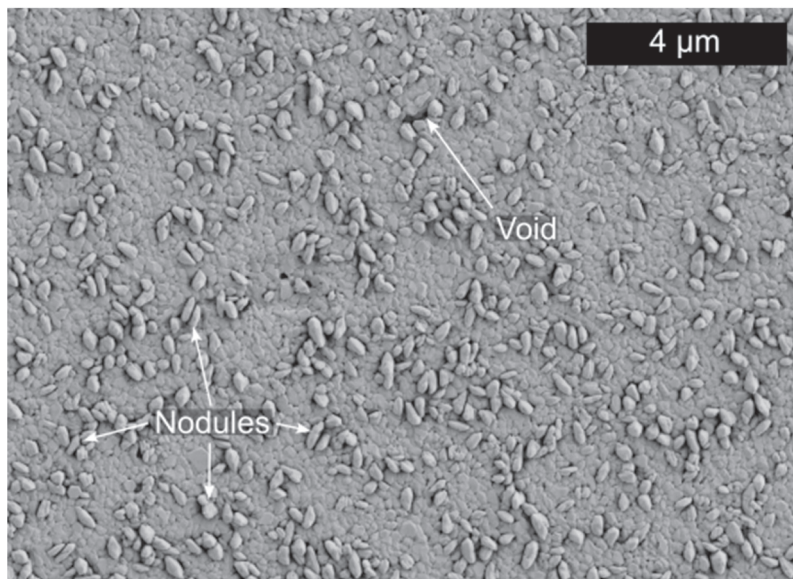


Figure 3. SEM images of vapor deposited TATB films grown on an BK7 glass substrate. (a) A cross-section of a TATB film showing columnar crystals, with nodules scattered on the free surface, and a sharp TATB/substrate interface. (b) Top-down view of the free surface showing the tile-pattern formed by the tops of the columns, the nodule defects, and voids (large dark areas).

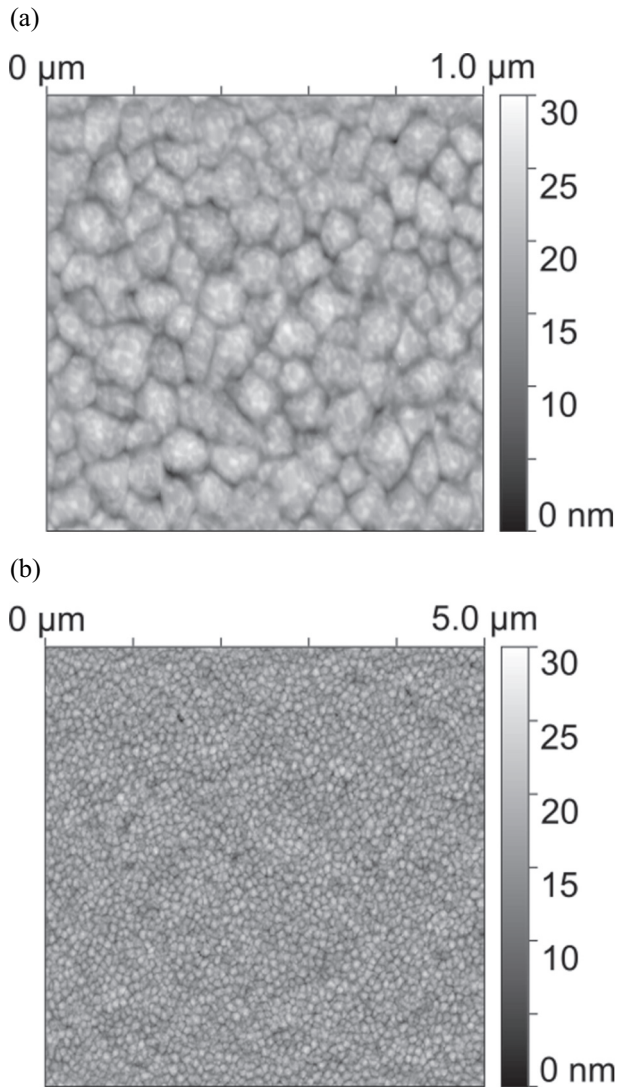


Figure 4. AFM images of the smooth TATB interface, with roughness variations forming a tiled polygonal pattern. Each polygon likely represents a separate TATB crystal grain. There are no noticeable changes to the pattern when zoomed in or zoomed out. (a) 1 μm × 1 μm scan with rms roughness 3.2 nm. (b) 5 μm × 5 μm scan of the same area also with rms roughness 3.2 nm. In total, scans were carried out in 3 different areas of the same TATB sample.

The roughness and topography of the buried TATB interface was measured with AFM using a TATB film liftoff sample (Figure 4). This interface had a mean rms roughness of 3.4 ± 0.1 nm. The AFM height scan revealed many polygonal domains forming a “tiled crystallite” pattern like that seen in previous work (Williamson et al. 2017). This pattern extends across the whole surface, seems to be the only significant topography pattern, and corresponds to the bottom of the columns observed by

Table 1. Thermal cycling for the two model PBX 9502 samples for NR: PBX1 and PBX2.

Sample	PBX1	PBX2
No Thermal Treatment	As-fabricated	As-fabricated
Thermal Treatment 1	Anneal 12 hours at 60°C	Anneal 12 hours at 60°C
Thermal Treatment 2	4 cycles between 25°C and 85°C, holding two hours at maximum temperature	Anneal 6 hours at 115°C
Thermal Treatment 3	4 cycles between 25°C and 85°C, holding two hours at maximum temperature	None

SEM (Figure 4). The width of these domains ranges from 50–200 nm. The lower end of this range aligns with the Scherrer crystalline domain size from XRD.

Neutron Reflectometry

Two different model PBX 9502 samples, PBX1 and PBX2, were measured before and after thermal cycling with NR. PBX1 and PBX2 were each measured several times: (1) in their initial state, (2) after a 12-hour 60°C anneal, and (3) after unique rounds of thermal cycling (Table 1). The unique thermal treatments were chosen to approximate typical PBX 9502 ratchet growth experiments to examine the relative importance of temperature cycling versus annealing time, but adapted to the highly compressed beamtime. 60°C was chosen for the initial anneal to bring Kel-F slightly above its glass transition temperature ($T_g = 27\text{--}32^\circ\text{C}$). For PBX1, the 85°C maximum temperature was chosen because it is the lowest temperature commonly used in ratchet growth experiments (Thompson et al. 2015; Thompson, Woznick, and DeLuca 2019; Woznick, Graff Thompson, and De Luca 2020). In general, a minimum ΔT of 60°C is needed to see significant ratchet growth. For PBX2, the 115°C maximum temperature was chosen because it is the lowest common ratchet growth temperature that is fully above the Kel-F melting point $T_m \approx 80\text{--}104^\circ\text{C}$ (Thompson et al. 2015; Thompson, Woznick, and DeLuca 2019). In many cases, thin-film samples delaminate or crack when thermally cycled. However, neither model PBX 9502 sample displayed damage when inspected visually. In addition, there were no visual changes or new features observed in the samples after thermal cycling. All the fitted parameters, their respective errors, scattering schematics, and fits to reflectivity data for PBX1 and PBX2 are presented in SI Section 6. Note, due to the low signal to noise in the NR measurements and the small q range because of limited beam time (see SI Section 6), the fitted roughness of the TATB – Kel-F interface and its increase with thermal cycling are not definitive (i.e., the maximum change in interfacial roughness is roughly within measurement error). Lower error measurements and a measurement range out to $q \approx 0.1 \text{ \AA}^{-1}$ are needed to be quantitative.

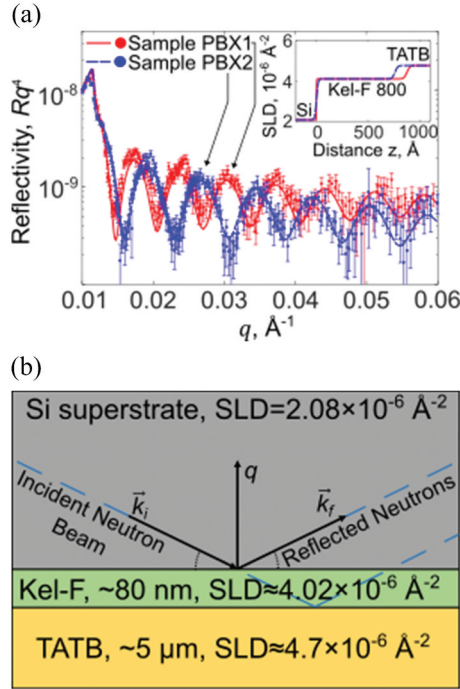


Figure 5. NR measurements for model PBX 9502 samples, PBX1 and PBX2, in their as-fabricated state. (a) Best fits to the reflectivity plots, where PBX1 was fit with $\chi^2 = 3.70$ and PBX2 was fit with $\chi^2 = 3.22$. Inset shows SLD profiles of the simulated NR best fits. (b) Schematic representing a model PBX 9502 sample and the path of reflected neutrons as they pass through it. Neutrons arrive at the sample as a collimated beam with wavevector $\vec{k}_i$ and travel through the side of the Si superstrate. The incident neutron beam is reflected off the Si – kel-F and TATB – kel-F interfaces. The reflected neutrons travel through the opposite side of the Si superstrate and to the detector with wavevector $\vec{k}_r$. The momentum transfer vector q is the vertical component of the difference between the incident and reflected wavevectors.

However, these experiments showed consistent trends in the modelled roughness of the TATB – Kel-F interface with thermal cycling and showed a clear difference in the reflectivity profile before and after cycling (SI Section 7). Importantly, methods to fabricate sufficiently smooth TATB samples that can be used to probe the (001) TATB – polymer binder interface have been demonstrated.

Before any thermal treatments (in their as-fabricated state), both PBX1 and PBX2 are similar (Figure 5a,b). Both samples have Kel-F and TATB layers that can fit to the same SLDs, with $\beta_{\text{Kel-F}} = 4.10 \pm 0.20 \times 10^{-6} \text{ \AA}^{-2}$ and $\beta_{\text{TATB}} = 4.76 \pm 0.13 \times 10^{-6} \text{ \AA}^{-2}$. These SLDs correspond to densities of $\rho_{\text{Kel-F}} = 2.06 \pm 0.10 \text{ g cm}^{-3}$ and $\rho_{\text{TATB}} = 1.88 \pm 0.05 \text{ g cm}^{-3}$ respectively ($\rho_{\text{Kel-F}}$ assumes a 0.25 molar fraction of VF₂ monomer). The SLDs of Kel-F and TATB for PBX1 and PBX2 agree with the Kel-F and TATB calibration samples.

The TATB – Kel-F interface roughness was about ~ 2.5 nm for both as-fabricated samples which is consistent with AFM measurements, Figure 4. The sharpness of the interface suggests there was almost no intermixing between Kel-F and TATB during sample fabrication. The thickness of the Kel-F layer is similar between PBX1 (88 ± 2 nm) and PBX2 (76 ± 2 nm), and the small difference is likely due to variation of the vapor saturated environment between rounds of spin coating. For both samples, the Kel-F – Si interface had low roughness, ~ 0.5 nm and ~ 1 nm for PBX1 and PBX2, respectively. This is expected for a spin coat. After these initial fits of PBX1 and PBX2 as fabricated, the Si, Kel-F, and TATB SLDs were kept constant when modeling

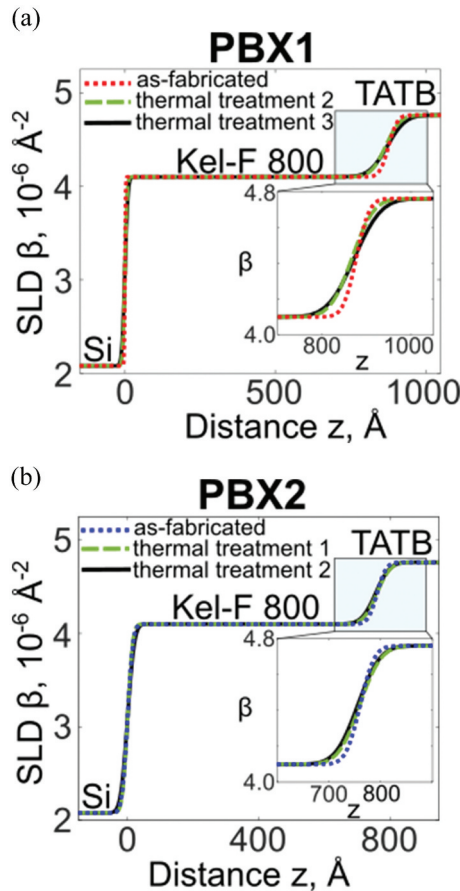


Figure 6. SLD profiles for the model PBX samples showing the area averaged SLD vs. The distance from the Si – kel-F interface. Roughness at the TATB – kel-F interface increases after thermal cycling. (a) SLD profiles for the best fits of PBX1. $\chi^2 = 3.70, 2.56$, and 2.58 for the as-fabricated, thermal treatment 2, and thermal treatment 3 fits, respectively. (b) SLD profiles for the simulated best fits of PBX2. $\chi^2 = 3.30, 3.79, 3.07$ for the as-fabricated, thermal treatment 1, and thermal treatment 2 fits, respectively. The observed changes in interfacial roughness are within measurement error, but the figure illustrates the trend of the TATB – kel-F interface toward greater roughness with thermal cycling.

thermal treatments to focus on changes happening at the TATB – Kel-F interface.

Both samples were subjected to an identical 60°C anneal for 12 hours (thermal treatment 1). PBX1 was not measured due to time constraints at the beamline. For PBX2, the TATB – Kel-F interface roughness fits to ~3.5 nm, which is an increase in roughness of ~1 nm. There was not any significant change in the roughness at the Si – Kel-F interface.

After the 60°C anneal, PBX1 and PBX2 were taken through different types of thermal treatment. Sample PBX1 was subjected to thermal treatments 2 and 3: both consisting of 4 rounds of temperature cycling between 25°C and 85°C. For each temperature cycle, the sample was held at 85°C for 2 hours. The goal was to observe changes to the interface under ratchet growth conditions. After thermal treatment 2, we observed several changes in PBX1 ([Figure 6a](#)). The modeled roughness of the TATB/Kel-F interface increased to ~4 nm. The roughening of the interface is suggestive of dewetting and creation of a depletion layer. After thermal treatment 3, the interface roughness further increased to ~5 nm.

After the 60°C anneal, PBX2 was subjected to another anneal at 115°C for 6 hours (thermal treatment 2). The goal of this anneal cycle was to observe changes to the interface when Kel-F was heated above its melting temperature. After the anneal cycle, the TATB – Kel-F interfacial roughness did not significantly increase ([Figure 6b](#)). This shows that heating the Kel-F above its melting point did not have a major effect on the TATB – Kel-F interface.

Discussion

TATB Film Microstructure

As detailed here, vapor deposited TATB films are an effective model for the TATB particles used in PBXs. Most importantly, vapor deposited TATB is the same crystalline polymorph as in PBX 9502. The (002) lattice reflection agrees with XRD measurements of normal TATB particles (Luscher et al. [2017](#); Schwarz et al. [2013](#); Yang et al. [2006](#)). The TATB thin film is highly textured (like neat-TATB and PBX 9502 pressed charges, but likely significantly more so), with SEM images and XRD scans indicating that the crystalline domains are well aligned.

This texturing likely results in the interfaces of the thin film sample being dominated by the (001) face, with the other main TATB crystal facets, the (100) and (01 $\bar{1}$) faces (Cady and Larson [1965](#)), being much less accessible. It is unclear whether this limits or improves the usefulness of the films. Firstly, the (001) face is by far the most prevalent TATB face in PBX 9502 due to the layered, sheet-like microstructure. Secondly, the PBX 9502 voids are aligned in

the same plane as the (001) face and are likely segregated to the TATB surfaces. Thus, void nucleation & growth within PBX 9502 is likely to primarily occur at the (001) TATB interface. Lastly, the (001) face is likely the most sensitive to interfacial aging phenomena. Molecular dynamic simulations predict a large surface energy difference between the (001) face and the other TATB crystalline faces (Gee et al. 2007). Although these simulations have major discrepancies with experiment (Bower, Kolb, and Pruneda 1980; Rivera and Matuszak 1983; Williamson et al. 2017; Yeager et al. 2010), the (001) face is still likely the lowest surface energy crystal face of TATB because it is the only face that includes the dispersive interaction contributions from the inner benzene ring. A lower TATB solid surface energy makes the (001) face more likely to experience debonding via decohesion and dewetting.

Decoherence, the separation of two solid surfaces by tensile or shear forces, is governed by the work of adhesion $W_{12} = (\gamma_1 + \gamma_2) - \gamma_{12}$, with surface energies γ_1 and γ_2 , and interfacial energy γ_{12} . The smaller W_{12} becomes, the easier it is to decohere the two materials. Thus, a TATB face with a significantly lower surface energy is more likely to decohere. Dewetting, the retraction of a liquid from a solid surface it was forced to cover, is governed by the spreading coefficient $S = W_{SL} - W_{LL}$, with solid – liquid adhesion W_{SL} and self-adhesion for the liquid W_{LL} . This simplifies to $S = \gamma_s - \gamma_{SL} - \gamma_L$. When $S \geq 0$, spontaneous wetting occurs. When $S < 0$ partial dewetting is observed. In the case of PBX 9502, TATB acts as the solid phase and Kel-F as the liquid. Although Kel-F is not technically a liquid, when raised above its T_g ($>32^\circ\text{C}$) Kel-F becomes sufficiently mobile that it could dewet over long periods of time (Karapanagiotis and Gerberich 2005; Reiter 1992). Thus, Kel-F is likely to dewet a TATB face with a significantly lower surface energy. Measurements and calculations of W and S are extremely difficult. As a consequence, there is no agreement in W or S values calculated from previously reported surface energies (Gee et al. 2007; Rivera and Matuszak 1983; Yeager et al. 2010). The ability to directly measure the emergence of voids at the interface with NR would shed light on the role of debonding within PBX 9502.

Thin films fabricated by vapor deposition commonly have some degree of internal stress. XRD measurements of TATB crystallite powder, which should not have internal stress, consistently measure the (002) reflection at $2\theta \approx 28.30\text{--}28.36^\circ$ (Luscher et al. 2017; Sun et al. 2010; Xu, Style, and Dufresne 2018; Yang et al. 2006). It was shown by Yeager et al. that pressing the neat TATB powder into a charge imparts strain into the TATB crystal that slightly changes the lattice parameters and diffraction reflections with the (002) reflection moved to $2\theta \approx 28.9^\circ$, calculated from reported lattice parameters using CRYSTALSIM software (Yeager et al. 2016). Similarly, for a pressed PBX 9502 charge the (002) reflection was reported at $2\theta = 28.66^\circ$ (Schwarz et al. 2013). The location of the (002) reflection measured for the

TATB thin film ($2\theta = 28.55^\circ$) falls in-between the unstressed and stressed samples. It is thus likely that the TATB thin films have a moderate amount of lattice strain in their as-fabricated state, but it is not possible to fully quantify how the lattice has changed with only one TATB reflection.

The TATB film forms as thin crystalline columns growing perpendicular to the substrate and in the [001] direction. This is unlike the stacked, sheet-like layers common to pressed TATB charges or the TATB particles. The thin film columns are straight, generally parallel to each other, and do not show any extreme variation in their width. The columnar structure and the slightly-domed column tops are reminiscent of Zone-M, or “matchstick morphology,” that forms under conditions of low adatom mobility (Knepper and Messier 2001). However, it is not clear that organic films deposited using physical vapor deposition should follow the same trends seen in structure zone models designed for inorganic materials. The mesh of polygonal domains at the TATB interface was observed in both our scans and those by Williamson et al. (2017). The widths of the domains are also roughly equivalent. Williamson et al. speculated that each domain grew from a single nanocrystal at the earliest stages of TATB deposition. They did not report roughness values of their films, but we can compare the height ranges of their AFM scans to the scans of the buried interface reported in this paper. Both AFM and NR measurements for the as-fabricated model PBX 9502 samples show that the roughness of the buried TATB interface was approximately 2–3 nm. The AFM measured height variation (see SI section 4, Fig S5) for the buried interface of the films is significantly less than that in Williamson et al.

The roughness of the substrate facing TATB interface is dominated by the TATB deposition process, rather than the surface roughness of the underlying polymer film. The surface roughnesses of the PAA and Kel-F films were about 0.3 nm and 0.9 nm respectively, which are much less than the buried TATB interfaces of the AFM liftoff sample and the NR fitted roughness of either PBX1 or PBX2. This suggests that neither the liftoff process nor choice of underlying polymer significantly affected the roughness of the buried TATB interface. The roughness of this interface is primarily due to how TATB nanocrystal grains interact as they grow. AFM linear height profiles show that the interfaces between the nanocrystalline domains significantly increased the surface roughness (see SI Section 4 and Figure S5). This can also be seen in Figure 4 as the dark outlines surrounding each of the domains.

The density of the TATB thin films measured by NR ($1.88 \pm 0.05 \text{ g cm}^{-3}$) is $\sim 97\%$ of the TMD for TATB, which is similar to the densities of highly compacted neat-TATB charges. Similarly, the density of Kel-F measured by NR ($2.06 \pm 0.10 \text{ g cm}^{-3}$) is almost identical to its maximum density (commonly given as 2.02 g cm^{-3}). Kel-F spin coat thicknesses for the model PBX 9502 samples measured with NR ($88 \pm 2 \text{ nm}$ and $76 \pm 2 \text{ nm}$) are consistent with ellipsometry on a representative Kel-F spin coat (79 nm). Overall, the

fitted NR models for as-fabricated PBX1 and PBX2 closely conform to what is expected based on theory and calibration measurements. This illustrates that the NR model samples are a good analogue of PBX 9502 and can be used for accurate NR measurements of the TATB – Kel-F interface.

NR Thermal Cycling

A key result from this work is methods to fabricate sufficiently smooth TATB samples that can be used to probe the (001) TATB – polymer binder interface. The low SLD contrast between TATB and Kel-F (SLD $\sim 4.76 \times 10^{-6} \text{ \AA}^{-2}$ and $\sim 4.10 \times 10^{-6} \text{ \AA}^{-2}$ respectively) makes NR highly sensitive to formation of voids (dewetting) at the interface due to the low scattering length density of air. Continuation measurements with longer count times to improve signal to noise and a q range extending to $0.1 \text{ \AA}^{-1}$ to improve our sensitivity to roughness would improve the resolution and enable more quantitative measurements.

The NR results show that, before thermal treatment, the TATB – Kel-F interface roughness was approximately the same for both model PBX 9502 samples and that NR can very likely measure changes in the model PBX 9502 samples with thermal cycling. Modelling NR data shows the changes are primarily happening at the TATB – Kel-F interface. After finishing all thermal treatments, the TATB – Kel-F interface roughness likely increased more in the thermally cycled sample PBX1 than in the high-temperature annealed sample PBX2. This indicates that repeated thermal cycling creates more roughness at the interface than annealing. The increased roughness, likely due to a depletion layer on the Kel-F side of the interface, is consistent with voiding at the interface. These results correspond well to literature on ratchet growth. Previous experiments have shown that while annealing above the Kel-F melting temperature can trigger void expansion (Thompson et al. 2015), thermal cycling is a much larger contributor to voiding (Thompson et al. 2015; Thompson, Woznick, and DeLuca 2019).

For sample PBX1, temperature cycling between 25 and 85°C raises the Kel-F above its glass transition temperature, $T_g > 32^\circ\text{C}$, allowing diffusion. Changes to TATB's morphology are unlikely because of its high melting temperature, $T_m \approx 350^\circ\text{C}$, and crystallinity. Therefore, voiding at the interface occurs in the Kel-F layer due to diffusion. Additionally, the anisotropic thermal expansion of TATB causes large expansions and contractions in the vertical direction, i.e. normal to the substrate (Kolb and Rizzo 1979). For the in-plane directions, the TATB thermal expansion is much smaller than that of Kel-F above its T_g (Cady and Caley 1977; Kolb and Rizzo 1979). The high degree of thermal expansion/contraction in the vertical direction and the thermal expansion

mismatch for the in-plane directions likely stresses the TATB – Kel-F interface, which may cause the Kel-F to further dewet or decohere from the TATB surface.

Sample PBX2 was annealed at 115°C, which is above Kel-F's melting temperature, $T_m \approx 95^\circ\text{C}$, further increasing Kel-F mobility. However, without the added stress from thermal cycling no additional voiding was detected. This is likely because the films were sufficiently thick to preclude dewetting in six hours.

In future work, the smooth TATB samples and the ability of NR to measure roughening with thermal cycling suggest that this will be a promising approach for directly quantifying voiding at the TATB – binder interface.

Conclusion

Improved capabilities for characterizing the TATB – binder interface are an urgent need for modeling thermomechanical behavior in PBX 9502 and transitioning away from Kel-F to effective but more environmentally friendly polymer binders. Central to this effort is the development of model TATB samples that can be used for interfacial measurements such as contact mechanics and NR. Overall, the thin film TATB samples developed and characterized here have met the requirements for high accuracy interfacial measurements. NR measurements can resolve changes to the TATB – Kel-F interface with thermal cycling. Additional rounds of thermal cycling and longer data collection times are poised to enable future quantification of changes at the TATB – Kel-F interface such as voiding and ratchet growth, which would improve resolution of this thermomechanical debonding mechanism.

Note

1. TATB crystallites can also be produced by wet-amination to make “chlorine-free TATB” or through an acid recrystallization process to make “ultra-fine TATB.” The chlorine-free TATB crystallites are similar to normal TATB in morphology, flocculation, and particle size, but are mostly free of voids, have much smoother surfaces, and are free of chlorine impurities (Rizzo, Humphrey, and Kolb 1981). The ultra-fine TATB also makes crystallites with smooth surfaces largely free of porosity (Nandi et al. 2007). However, the crystals are bi-pyramidal and agglomerate into fractal-looking particles with a bi-modal size distribution ranging from ~100 nm to 10 μm .

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Data Availability Statement

The underlying data supporting the findings of this study are not publicly available and strict restrictions apply to their availability due to the nature of the research and U.S. Department of Energy policy. These data can be made available from the corresponding authors, TLK and RK, upon reasonable request.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the authors used ChatGPT-3o and –4o to improve the readability and conciseness of several portions. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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